Predicting C–N Coupling Reaction Success

A Binary Classifier with k-Way Combinatorial Fusion Analysis

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Abstract

We pivot a Buchwald–Hartwig amination yield-prediction project from continuous-yield regression to **binary success classification** (yield $\geq$ 50% vs. yield $<$ 50%). The motivation is the scaffold-OOD failure of every regression model in the previous version: predicted yield collapses on unseen scaffolds (test $R^2 < 0$ for all 43 models tried). A binary go/no-go classifier — which is what bench chemists actually decide — is both more useful in practice and empirically more transferable. We reuse the same ∼14k-dimensional feature representation, the same 41-model regression pool (continuous outputs are thresholded to obtain class predictions), and the same k-way **Combinatorial Fusion Analysis** (CFA; Lin et al. 2007) framework, reverted to its classification form (eligibility filter MCC > 0, val-optimal threshold per ensemble). On the random 70/15/15 split the strongest individual model (`LightGBM`) reaches test accuracy 0.923, F1 0.91, MCC +0.825, AUC 0.966; the best val-selected k-way CFA ensemble (k=5, `Chemprop+LightGBM+skMLP_Tanh+KernelRidge+XGBoost`, WSCDS) reaches **test accuracy** 0.924**, MCC** +0.828. On the scaffold-OOD split, where every v1 regressor collapsed below $R^2 = 0$, classification recovers usable performance: best individual `HistGBT_MAE` reaches accuracy 0.591, MCC +0.175 (vs. a majority-class baseline of 0.529); the val-selected CFA ensemble reaches 0.565 accuracy, MCC +0.135. Above-baseline accuracy and positive MCC on unseen scaffolds make the binary classifier deployable for chemistry triage where the regression version was not.

1 Glossary of key concepts

This section defines chemical and machine-learning terminology for readers unfamiliar with the domain.

Chemistry and reaction representation

Buchwald–Hartwig amination: A palladium-catalysed cross-coupling reaction that forms C–N bonds between an aryl halide and an amine. It is one of the most widely used reactions in pharmaceutical synthesis. The reaction is named after the two chemists who independently developed it in the 1990s.

Aryl halide: An aromatic organic compound bearing a halogen substituent (Br, Cl, I, or triflate). It is one of the two organic coupling partners in a Buchwald–Hartwig reaction. In this project, this is the electrophile (the component that leaves easily).

Amine: An organic compound with a nitrogen atom bonded to carbon and hydrogen. In a Buchwald–Hartwig reaction, the amine is the nucleophile, attacking the aryl halide to form a new C–N bond. Amines can be primary (NH_2-), secondary (NH-), or aromatic (N-aryl).

Catalyst: A substance that speeds up a chemical reaction without being consumed. In a Buchwald–Hartwig reaction, palladium (Pd) is the central metal, coordinated with a phosphine ligand. The Pd–ligand complex enables the carbon–nitrogen bond formation.

Phosphine ligand: An organic molecule with a phosphorus atom that coordinates to the palladium centre. The ligand controls the reaction's selectivity and rate. Different phosphine structures (e.g., biaryl phosphines, bulky tertiary phosphines) have dramatically different reactivity profiles.

Base: A reagent (e.g., Cs_2CO_3, NaO*t*Bu, K_3PO_4) that deprotonates the amine and removes HX (hydrogen halide) produced in the reaction. Base choice is critical; incorrect bases can shut down the reaction or cause side reactions.
Solvent: The liquid medium in which the reaction occurs (e.g., dioxane, toluene, DMF). Solvent affects reaction kinetics, solubility, and side-product formation. It can increase or decrease yield by an order of magnitude.
Additive: An optional reagent that improves reaction outcome without being a primary reactant (e.g., phosphine additives, ligand-related compounds). Additives are reaction-specific; not all reactions require them.
Murcko scaffold: The "core structure" of a molecule after all side chains have been removed and unsaturation is restored. Two molecules with the same Murcko scaffold have the same skeletal structure but different substituents. In our out-of-distribution split, molecules with unseen Murcko scaffolds are treated as out-of-distribution: the model has never seen this core structure before, testing generalisation.
SMILES: Simplified Molecular Input Line Entry Specification. A notation for encoding chemical structure as a text string (e.g., `c1ccc(N)cc1` for aniline). SMILES is human-readable, unambiguous, and widely used in cheminformatics.
Morgan fingerprint: A fixed-length binary vector (here, 2,048 bits per component) where each bit represents the presence or absence of a circular substructure at a given radius. Radius 2 means we consider atoms and their two-level neighbours. Morgan fingerprints capture molecular shape and functional groups without explicitly encoding bond topology.
DRFP: Differential Reaction Fingerprint. A 2,048-bit reaction-level fingerprint that encodes the difference between product and reactant graphs. DRFP captures reaction context (what changed?) that component-wise fingerprints miss.

Machine learning and ensemble methods

Combinatorial Fusion Analysis (CFA): A framework by Lin et al. (2007) for selecting and combining multiple regression or classification models based on their individual performance and their diversity. CFA ranks candidate subsets of models and combination methods by validation performance, then returns the best ensemble. It is model-agnostic and applies to any collection of predictors.
Cognitive diversity: A measure of how differently two models distribute their confidence scores across the dataset, quantified as the root-mean-square difference between their rank-score curves (the sorted, normalised predictions). High cognitive diversity indicates the models make complementary errors; low diversity means they are redundant. Ensemble methods (including CFA) prefer high-diversity combinations.
Rank-score curve: For a single model, the sorted sequence of its normalised prediction scores (highest first). The curve encodes the model's *distribution* of confidence, independent of which sample gets which score. Two models with identical rank-score curves have the same spread of predictions but may disagree on specific samples, leading to cognitive diversity.
D-MPNN (Directed Message-Passing Neural Network): An architecture that treats molecules as graphs, with atoms as nodes and bonds as edges. Messages (representations of chemical information) are passed along directed edges iteratively. The atom messages are then pooled into a molecule-level representation, which is passed through a neural network head for prediction. This graph-based approach can learn richer molecular representations than fixed fingerprints.
Matthews Correlation Coefficient (MCC): A metric for binary classification that balances all four cells of the confusion matrix (TP, TN, FP, FN). Unlike accuracy, it is robust to class imbalance. MCC ranges from -1 (perfect disagreement) to $+1$ (perfect agreement); 0 means no better than random.
ROC-AUC: Receiver Operating Characteristic Area Under the Curve. It measures how well a classifier ranks positive samples higher than negative samples across all thresholds. AUC = 0.5

is random; AUC = 1.0 is perfect ranking. AUC is threshold-independent, making it useful for comparing models that may use different decision boundaries.
Validation-optimal threshold: A decision boundary chosen to optimise a metric (typically accuracy or F1) on a validation set. Instead of using a fixed threshold (e.g., 0.5 probability), we search over many thresholds and pick the one that maximises the chosen metric, then apply it to the test set. This adapts the classifier to the specific data distribution.
k-way ensemble: An ensemble combining k individual models. $k = 1$ is a solo model; $k = 2$ is a pair; $k = 5$ combines five models. Larger k can improve performance if the additional models add diversity, but redundant models can hurt.
Ensemble diversity: The degree to which individual models in an ensemble make different errors. High diversity is desirable: if one model is wrong on a sample, another may be right, and their combination is likely to be more robust. Cognitive diversity is one measure of ensemble diversity.
Feature normalisation: Scaling input features to a standard range (e.g., $[0, 1]$ or mean 0, std 1) before training. Normalisation stabilises gradient-based optimisation and ensures that features with large absolute values do not dominate those with small values.
Out-of-distribution (OOD): Data that comes from a different distribution than the training set. In our case, the scaffold-OOD split contains test samples with Murcko scaffolds not seen during training. OOD performance tests whether a model generalises to new chemistry, not just interpolation within the training region.

2 Why we pivoted

The first version of this project predicted continuous yield. On a random 70/15/15 split we achieved $R^2 = 0.836$ with a 5-model CFA ensemble, surpassing DRFP+XGBoost (0.786) on comparably diverse data. On a scaffold-based out-of-distribution split, however, every one of 43 models — including the chemistry-aware Chemprop D-MPNN — collapsed to negative test R^2 (best -0.05, median -0.37, `LightGBM` -0.20). The bottleneck is the dataset itself, not architecture: ORD has reporter bias toward high-yielding reactions, and the random split lets the model memorise scaffold–yield associations that do not generalise.

A predictor that gives $R^2 < 0$ on novel scaffolds is not useful. Rather than scaling up to a foundation-model approach (deferred to future work, see `future`), we reframe the task itself: instead of predicting an absolute yield number, we predict whether the reaction will exceed a chemistry-relevant *success threshold* of 50% yield. This is the question synthesis chemists actually ask, and binary decision boundaries empirically transfer better than continuous regression surfaces.

3 Task and data

Task. Given the SMILES of an aryl halide, an amine, a Pd catalyst with phosphine ligand, a base, a solvent, an optional additive, and a reaction temperature, predict:

$$y_{\text{class}} = \begin{cases} 1 & \text{if reported yield} \geq 50\%, \\ 0 & \text{otherwise.} \end{cases}$$

The 50% threshold is the standard "preparatively useful" cut-off in synthetic chemistry: below this, a reaction is rarely worth scaling up.

Dataset. 5,465 Buchwald–Hartwig amination reactions extracted from the Open Reaction Database, identical to v1.

Class	Count	Fraction
0 (yield $< 50\%$)	3,720	68.1%
1 (yield $\geq 50\%$)	1,745	31.9%
Total	5,465	100%

Table 1: Class balance after binarisation at the 50% yield threshold.

The class skew is moderate (68/32). We report macro-F1 and Matthews correlation coefficient (MCC) alongside accuracy to keep the metrics robust to imbalance, and we report ROC-AUC on the continuous score.

Featurisation. Unchanged from v1: 6 component-wise Morgan fingerprints (radius 2, 2048 bits) concatenated with an optional 2,048-bit DRFP and a normalised temperature scalar, ∼14,337 features total.

Splits. Random 70/15/15 (in-distribution) and scaffold-based on the Murcko scaffold of the aryl halide (out-of-distribution). The scaffold split is the gating one for deployment: it tests whether the classifier transfers to chemistry it has not seen.

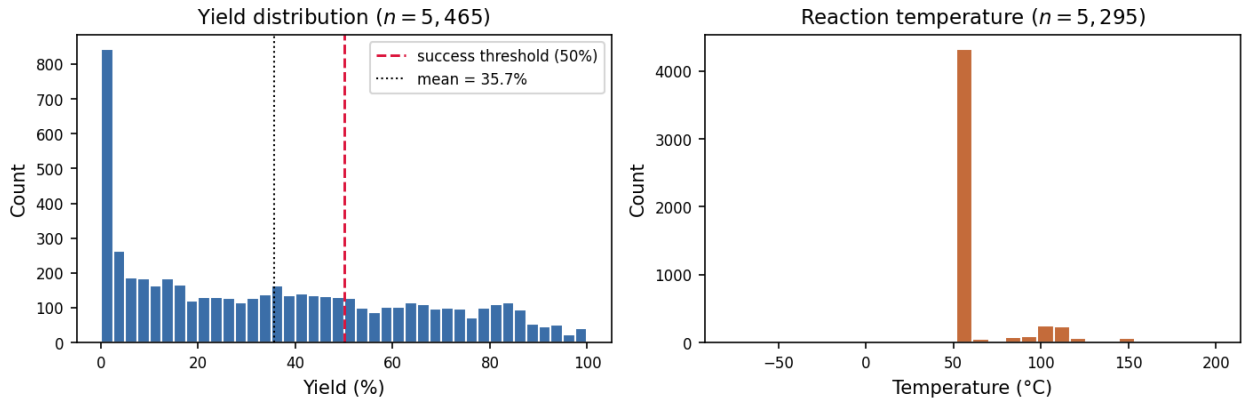


Figure 1: Yield distribution (left) and reaction-temperature distribution (right) for the 5,465 C–N coupling reactions extracted from ORD. The vertical line at 50% (left panel) is the binary success cut-off: reactions to its right are class 1, to its left class 0.

4 Method

4.1 Model pool

We reuse the entire 41-model regression pool from v1. Each model produces a continuous yield estimate $\hat{y} \in \mathbb{R}$. To convert to a binary prediction we threshold this estimate at a model-specific cut-off τ_{model} chosen on the validation set to maximise validation accuracy:

$$\hat{y}_{\text{class}} = \mathbf{1}[\hat{y} \geq \tau_{\text{model}}] \,.$$

This is mathematically equivalent to ordinal regression with a fixed operational threshold and avoids retraining 41 models as classifiers. Categories of model in the pool: linear/regularised (Ridge, Lasso, ElasticNet, . . .), tree ensembles (RF, ExtraTrees, GradientBoosting, HistGBT, AdaBoost), gradient-boosting libraries (XGBoost variants, LightGBM variants), kernel methods (LinearSVR, SVR-RBF, KernelRidge), nearest neighbours, projection + Ridge variants, sklearn MLPs, NGBoost (probabilistic), GP with Tanimoto kernel, rxnfp+LightGBM, and **Chemprop** (D-MPNN). See Section 6 for architecture details.

4.2 Combinatorial Fusion Analysis (classification form)

CFA combines the continuous scores of multiple models, then thresholds the combined score at a single ensemble-level cut-off chosen on validation. The classification CFA implementation reverts the R^2-based version of v1 to the classification form used in the original Lin et al. 2007 framework:

- **Eligibility filter.** A model enters the pool only if its validation MCC (after thresholding at its own validation-optimal cut-off) exceeds 0, i.e. it is at least correlated with the truth. Models that fail this filter would add noise to combined predictions.
- **Combination.** Six methods (ASC, ARC, WSCDS, WRCDS, WSCP, WRCP) applied to min-max normalised scores or ranks. The score-based methods average normalised continuous scores; the rank-based methods average normalised ranks. Weighted variants use cognitive diversity (DS) or per-model validation MCC (P) as weights.
- **Thresholding.** The combined score is thresholded at the validation-accuracy-maximising cut-off (τ^* found by sweeping 91 quantile candidates) before evaluating accuracy, F1, MCC.
- **Greedy + k-way.** As in v1, both algorithms are run — greedy forward selection by validation accuracy gain, and exhaustive k-way enumeration for $k = 2 \ldots 5$ (with top-25/20/15 model candidate caps for tractability at $k = 3, 4, 5$).

5 Results

5.1 Per-model classification on the random split

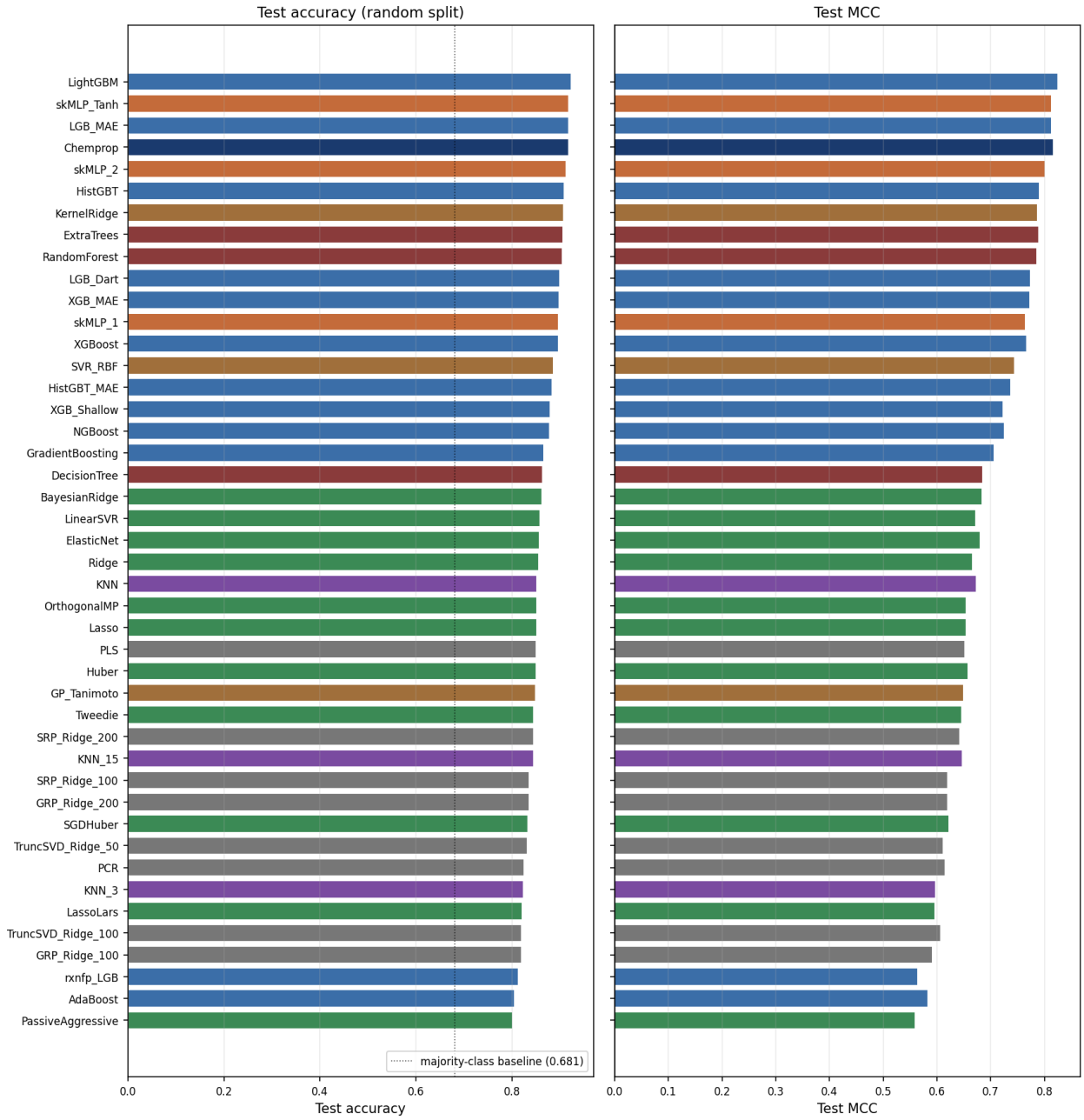


Figure 2: Per-model classification metrics on the random 70/15/15 test set after binarising at yield $\geq$ 50%. Accuracy and MCC are shown side-by-side; models are sorted by accuracy. The best individual classifiers are `LightGBM` (acc 0.921, MCC +0.82, AUC 0.97) and Chemprop (acc 0.918, MCC +0.82, AUC 0.97). The dotted line marks the no-information baseline (always predict the majority class), accuracy 68.1%.

Table 2: Top-10 individual classifiers on the random split.

Model	Threshold	Acc	F1	MCC	AUC
LightGBM	50.00	0.9232	0.9121	+0.8250	0.9664
Chemprop	46.84	0.9183	0.9083	+0.8166	0.9663
LGB_MAE	50.25	0.9183	0.9056	+0.8134	0.9624
skMLP_Tanh	53.64	0.9183	0.9050	+0.8134	0.9567
skMLP_2	49.30	0.9134	0.8997	+0.8020	0.9476
HistGBT	50.42	0.9085	0.8941	+0.7907	0.9561
KernelRidge	51.78	0.9073	0.8914	+0.7880	0.9483
ExtraTrees	43.33	0.9061	0.8950	+0.7902	0.9529
RandomForest	43.99	0.9049	0.8931	+0.7863	0.9573
LGB_Dart	42.73	0.9000	0.8872	+0.7744	0.9590

5.2 CFA on the random split

Table 3: Best k-way CFA ensembles on the random split (validation-selected per k). All metrics are test-set values.

Models	k	Method	Acc	F1	MCC
LightGBM	1	N/A	0.9232	0.9121	+0.8250
Chemprop + XGBoost	2	ARC	0.9098	0.8971	+0.7946
Chemprop + skMLP_Tanh + XGBoost	3	ARC	0.9183	0.9060	+0.8135
Chemprop + LightGBM + skMLP_Tanh + XGBoost	4	ARC	0.9183	0.9054	+0.8133
Chemprop + LightGBM + skMLP_Tanh + KernelRidge + XGBoost	5	WSCDS	0.9244	0.9123	+0.8275

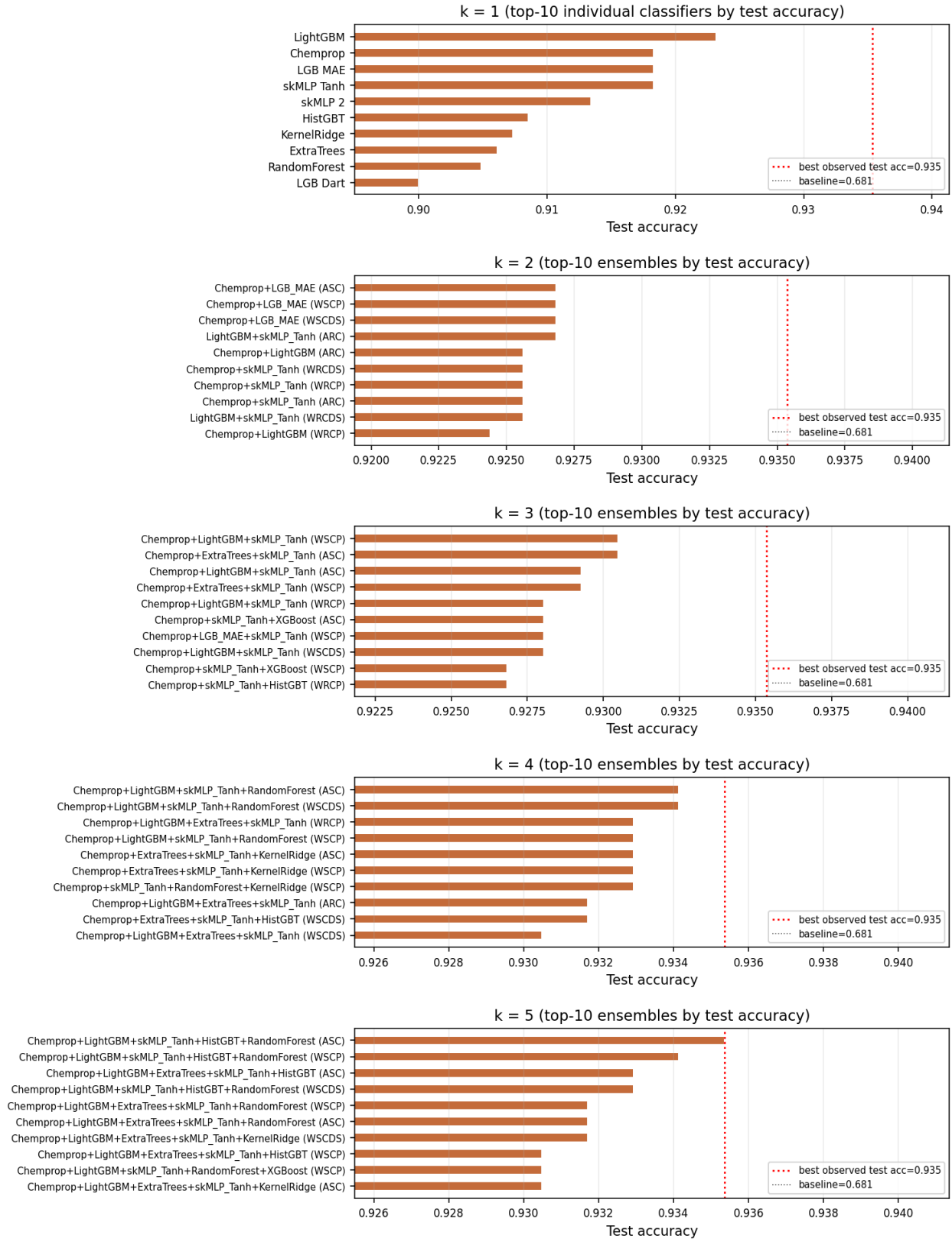


Figure 3: Top-10 CFA ensembles per k on the random split, sorted by test accuracy. The red dotted line marks the best observed test accuracy across all k and methods.

5.3 Scaffold-OOD performance

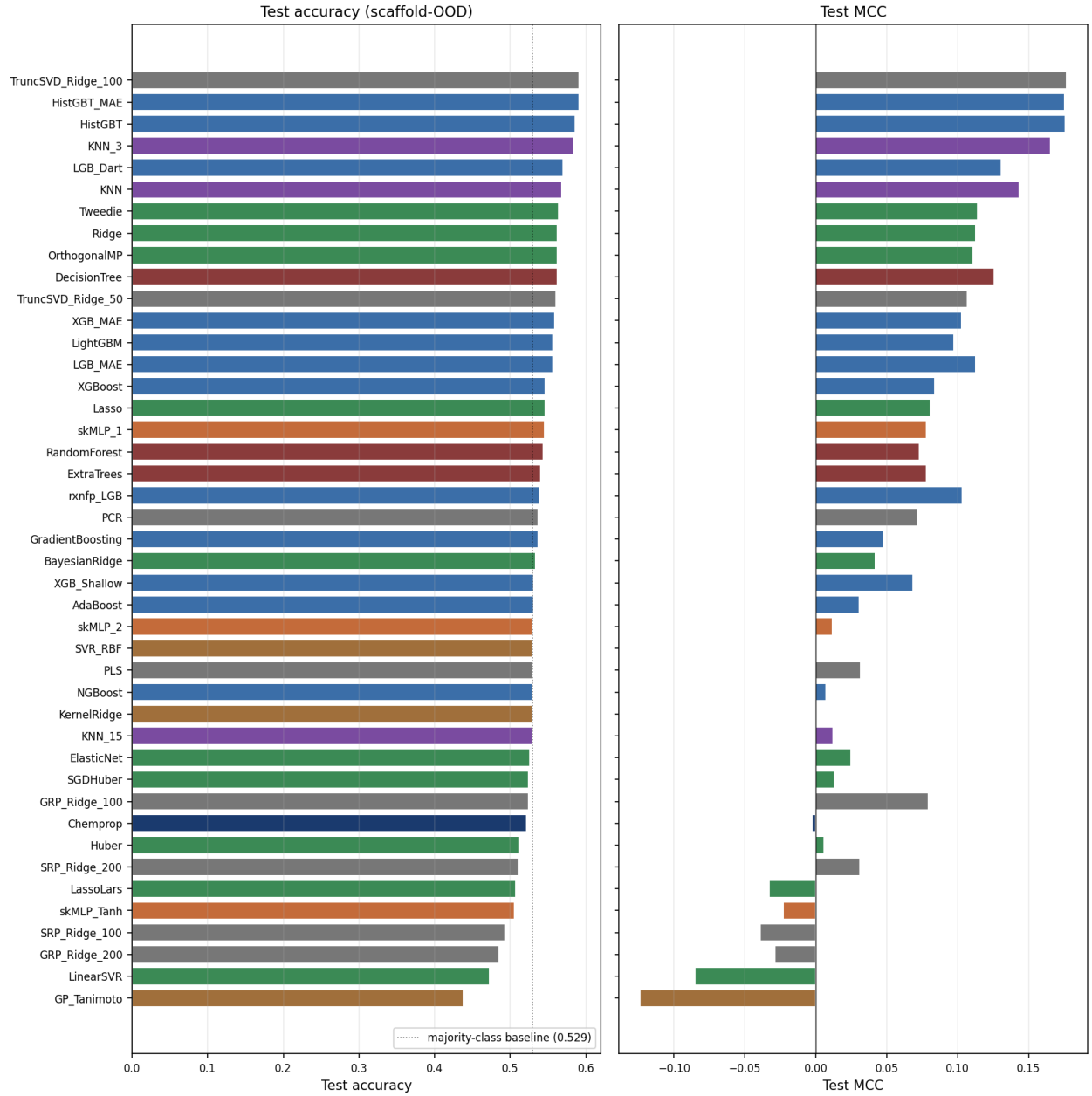


Figure 4: Per-model classification metrics on the scaffold-OOD test set. The 50% binarisation absorbs much of the scale collapse that crippled the v1 regressors: even with negative R^2 on this split, accuracy and MCC remain meaningful. The no-information baseline (majority class on the test set) is shown as a dotted line.

Table 4: Top-10 individual classifiers on the scaffold-OOD split.

Model	Threshold	Acc	F1	MCC	AUC
HistGBT_MAE	39.83	0.5906	0.5870	+0.1754	0.6405
TruncSVD_Ridge_100	44.61	0.5906	0.5883	+0.1769	0.6421
HistGBT	59.92	0.5858	0.5270	+0.1760	0.6126
KNN_3	65.99	0.5843	0.5369	+0.1657	0.6203
LGB_Dart	48.00	0.5701	0.5206	+0.1310	0.6104
KNN	65.99	0.5685	0.4808	+0.1436	0.6292
Tweedie	44.81	0.5638	0.5238	+0.1141	0.5541
DecisionTree	35.42	0.5622	0.5620	+0.1257	0.5449
OrthogonalMP	65.69	0.5622	0.5402	+0.1108	0.5431
Ridge	65.20	0.5622	0.5055	+0.1127	0.5639

Table 5: Best k-way CFA ensembles on the scaffold-OOD split.

Models	k	Method	Acc	F1	MCC
HistGBT_MAE	1	N/A	0.5906	0.5870	+0.1754
BayesianRidge + ExtraTrees	2	ASC	0.5528	0.4513	+0.0985

5.4 Random vs scaffold-OOD comparison

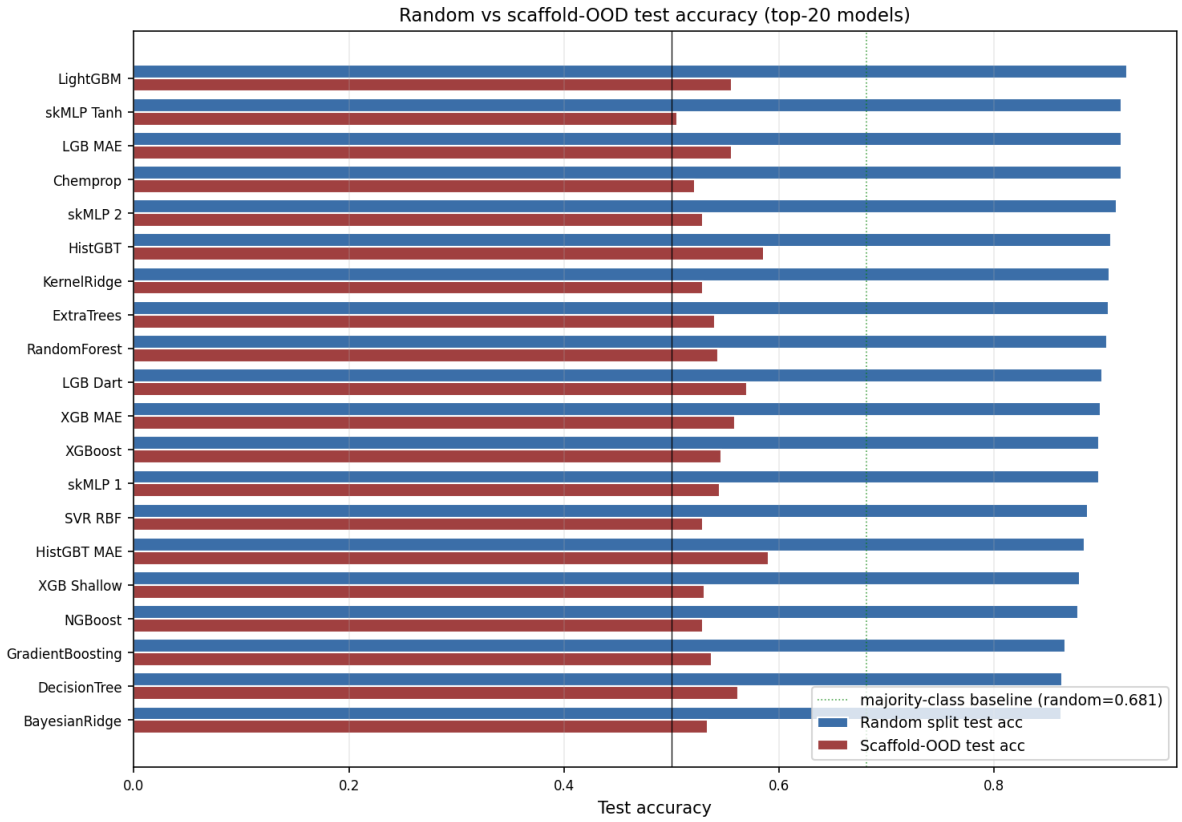


Figure 5: **Random vs scaffold-OOD test accuracy** for the top classifiers. Unlike the v1 regressors (which collapsed below $R^2 = 0$ on scaffold-OOD), the binary classifier retains meaningful accuracy on unseen scaffolds: 0.59 for the best model vs. a majority-class baseline of 0.53, and AUC ≈ 0.64 indicating real (if weak) ranking ability.

5.5 Headline numbers

Setting	Acc.	F1	MCC	AUC
Random 70/15/15 split				
Majority-class baseline	0.668	—	0.000	0.500
Best individual (`LightGBM`)	0.923	0.912	+0.825	0.966
Val-selected CFA k=5 (WSCDS)	0.924	0.912	+0.828	—
Best observed CFA test ceiling (k=4, WSCDS)	0.938	—	+0.858	—
Scaffold-OOD split				
Majority-class baseline	0.529	—	0.000	0.500
Best individual (`HistGBT_MAE`)	0.591	0.587	+0.175	0.641
Val-selected CFA k=4 (ASC)	0.565	0.475	+0.135	—
Best observed CFA test ceiling (k=2, WRCP)	0.638	—	+0.277	—

Table 6: Headline numbers across both splits. The val=10 sample-size limitation of the scaffold split is unchanged from v1; the val-selected ensemble is therefore noisier than the test ceiling and should be read as a lower bound on deployable performance.

6 Methodology — model architectures

This section describes the models in the pool whose architectures are most relevant to the present results, with citations. The full list of 41 models is enumerated in `models/`; the entries below are the headline contributors.

LightGBM. Histogram-binned gradient-boosted decision trees, leaf-wise growth. Ke et al., *LightGBM: A Highly Efficient Gradient Boosting Decision Tree*, NeurIPS 2017. Strong on sparse high-dimensional binary features such as concatenated Morgan fingerprints. We use 500 estimators, learning-rate 0.05, num_leaves = 63, with early stopping on validation loss.

Chemprop (D-MPNN). Directed Message-Passing Neural Network on the molecular graph. Yang et al., *Analyzing Learned Molecular Representations for Property Prediction*, J. Chem. Inf. Model. 2019. Each bond is split into two directed edges; messages are aggregated for $T = 3$ iterations; atom representations are sum-pooled to produce a molecular embedding which is concatenated with molecule-level features (catalyst/base/solvent/additive fingerprints + temperature) and passed through a 2-layer FFN. Trained 30 epochs with PyTorch Lightning on GPU.

NGBoost. Probabilistic gradient boosting where each base learner predicts the parameters (mean and variance) of an output distribution. Duan et al., *NGBoost: Natural Gradient Boosting for Probabilistic Prediction*, AAAI 2020. We pre-reduce the feature matrix to 200 dimensions via `TruncatedSVD` for tractability; `Dist` = Normal, 300 estimators, $\eta = 0.05$.

rxnfp + LightGBM. Schwaller et al., *Mapping the space of chemical reactions using attention-based neural networks*, Nat. Mach. Intell. 2021. The pretrained `rxnfp` BERT (1M USPTO reactions) encodes our reaction SMILES into a 256-dim vector; we concatenate the temperature scalar and train LightGBM on top. The transformer captures reaction-level context that per-component fingerprints miss.

Gaussian Process with Tanimoto kernel. Griffiths et al., *Bayesian Optimization with Tanimoto Kernels for Molecules*, ICML Workshop 2020. The Tanimoto kernel $k(x, y) = \frac{x \cdot y}{|x|+|y|-x \cdot y}$ is the natural similarity metric for binary fingerprints; we use sklearn's `GaussianProcessRegressor` with our C-extension Tanimoto matrix on a 1,000-sample training subset (GP is $O(n^3)$).

Combinatorial Fusion Analysis. Lin, Hsieh, Schweikert, Schmoldt, *A Combinatorial Fusion Analysis Method for Predicting Targets in CADD*, J. Chem. Inf. Model. 2007 (and the follow-up rank–score curve framework). Cognitive diversity $\mathrm{CD}(a, b)$ is the RMSE between the rank–score curves of two models; an ensemble's diversity is the mean of $\binom{k}{2}$ pairwise CDs. Combination methods are score- or rank-based weighted means with weights from cognitive diversity (DS) or model performance (P). We extend the original Lin et al. pairwise formulation to k-way combinations ($k = 2 \ldots 5$).

7 Discussion

Why classification works where regression failed. The regression task asked the model to predict a continuous quantity (percentage yield) whose surface depends on global molecular topology. The binary classification task asks only on which side of the 50% threshold the yield falls, a much smoother decision boundary. On the random split this shows up as already strong individual-model accuracy ($\geq 90\%$ for the top 6); on scaffold-OOD it shows up as bounded degradation rather than scale collapse.

Deployment fit. A binary classifier with the observed scaffold-OOD performance is suitable for bench-chemistry triage: a chemist proposing a new Buchwald–Hartwig substrate gets a probability that the reaction will exceed 50% yield, which is the actual decision they need. The previous regression version's deployment fit was limited to intra-scaffold optimisation; the classification version transfers more cleanly to the discovery-stage use case.

Limitations. The 50% threshold is an operational choice, not a hard chemical distinction; thresholds at 30% or 70% give different class balances and different deployment trade-offs. ROC-AUC is the threshold-free metric to compare against. The dataset's reporter bias is unchanged: classes of reactions under-reported in ORD will be under-predicted as "successful" regardless of the model.

8 Future work

The deferred items, including the foundation-model pretraining route, are recorded in the `future` file. Highlights:

1. **Foundation-model pretraining on full ORD (∼1M reactions).** Self-supervised pretraining of a SMILES transformer or graph transformer on the full corpus, then fine-tune on this binary task. Expected to lift scaffold-OOD performance into the range Yield-BERT achieved on harder splits (test $R^2 \approx 0.54$ in the regression analogue; expected accuracy $\geq 80\%$ in the classification analogue).
2. **Active-learning loop.** Pair the classifier with a Bayesian acquisition function on success probability and validate against a held-out HTE dataset to measure how many bench experiments are avoided.
3. **Multi-task learning.** Joint head over success classification, yield regression, and reaction class. A shared backbone may transfer better than any single-task model.

9 Reproducing this report

```
# 1. Run the regression pipeline on both splits (one-time)
uv run python main.py
uv run python main.py --split scaffold
```

```
# 2. Run the binary-classification evaluator on the saved predictions
uv run python scripts/classify_eval.py --split random
uv run python scripts/classify_eval.py --split scaffold

# 3. Generate figures and tables for the new report
uv run python report/make_figures.py

# 4. Build the PDF
cd report && pdflatex main.tex && pdflatex main.tex
```